

Финограф — Methodology Verification Report

Cash-flow and debt-health conformance testing of the questionnaire prototypes (full & limited versions)

Prepared: 9 August 2026 · Reference source: methodology-source.xlsx (analyst specification)

Summary

Two independent, deterministic test suites were run against the questionnaire prototypes to check whether the answers a user provides produce the scores defined by the analyst's methodology spreadsheets. Findings:

- **Cash-flow model ("Методика/Отчет паричен поток"):** **fully conforms**. All six test profiles reproduce the methodology to the decimal — component percentages, component points and the weighted overall cash-flow score all match exactly.
- **Debt-health model ("Методика/Отчет дългово здраве"):** **partial (free-tier subset)**. The prototype implements only three of the five methodology components (and two of those with simplified inputs). It therefore diverges from the full methodology — in the tested risk cases it **overstates** debt health by up to ~44 points. This is by design: the full debt engine is scoped as paid-tier work that is not yet built.
- The **full** and **limited** prototypes share one identical scoring engine; both were run and produced identical results. They differ only in what is shown after scoring (full report vs. locked/sample result).

1. What was tested

File	Role	Questionnaire	Scoring engine
prototype-full.html	Paid — full report	Full 85-field questionnaire	Shared engine
prototype-limited.html	Free — locked / sample result	Full 85-field questionnaire	Shared engine (identical output)

Both versions were verified against two areas of the analyst specification: the cash-flow sheets and the debt-health sheets.

2. Method

For each area, the rules from the spreadsheet were re-implemented **independently** in a separate program (a second, clean interpretation of the analyst's formulas and scoring bands). Fixed answer combinations ("profiles") were then run through the prototype's **actual** scoring code, and every value was compared field-by-field against the independent implementation. The tests are headless and deterministic, and are provided as two reproducible scripts (see section 6). This cross-check catches both formula errors and input-mapping errors.

3. Cash-flow results — CONFORMS

Methodology basis (from the sheet): net income = sum of income fields; essential expenses = housing/utilities/food/transport/health/dependants/internet + debt payments + insurance + annualised irregular costs; non-essential = discretionary spending; savings = monthly savings + voluntary pension. Overall = 40% essential + 20% non-essential + 40% savings, each scored on the sheet's piecewise bands.

Profile	Essential / Non-ess / Savings	Points (ess / non / save)	Overall	Result
P1 Balanced	40% / 20% / 30%	100 / 100 / 100	100	MATCH
P2 Good / mid	55% / 30% / 12%	70 / 80 / 48	63.2	MATCH
P3 Deficit	75% / 45% / 0%	20.8 / 36.7 / 0	15.7	MATCH
P4 Debt-heavy	73.3% / 10% / 13.3%	22.2 / 100 / 53.3	50.2	MATCH
P5 Annual fields (÷12)	40% / 15% / 15%	100 / 100 / 60	84	MATCH

P6 Multi-income	40% / 20% / 24%	100 / 100 / 88	95.2	MATCH
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In every profile the prototype and the methodology produce the same numbers. The set covers the edge cases: mortgage/debt folded into essential, annual fields divided by 12, multiple income sources summed, a monthly deficit, and each scoring-band boundary.

4. Debt-health results — PARTIAL (free-tier subset)

The methodology defines five weighted components. The prototype implements a simplified subset and re-weights it, so the debt score diverges from the full methodology. Results:

Profile	Methodology (final — level)	Prototype	Gap
D1 Clean / low debt	93.6 — Отлично	82.2	-11.4
D2 Big mortgage (€200k)	81.4 — Много добро	93.6	+12.2
D3 Expensive card, debt > assets	39 — Рисково	78.1	+39.1
D4 Current arrears > 30 days	19 — Критично	62.9	+43.9
D5 Overdraft dependence	86.8 — Много добро	96	+9.2

A positive gap means the prototype scores debt health **higher** (better) than the methodology. The prototype overstates in the risky cases (D3, D4) because it omits the cost-of-debt and card/overdraft components and does not apply the methodology's cap ladder to the debt score; and in D2 because it ignores the mortgage balance.

Component-by-component conformance

Component (weight)	Status	Detail
Payments / income (35%)	Conforms (curve)	Scoring curve matches the sheet exactly. Reads the aggregate payment fields (EXP-02 + EXP-11) rather than the per-contract fields (DEB-03/09/15).
Debt / assets (15%)	Partial	Formula matches, but inputs differ: the prototype excludes the mortgage balance from debt, and excludes property, pension, business and personal assets from assets.
Cost of debt (15%)	Not implemented	No comparison of contract interest rates against the BNB market benchmarks.
Cards & overdraft (10%)	Not implemented	No card utilisation, repayment-behaviour or overdraft-usage scoring.
Discipline + ЦКР (25%)	Partial	Uses the missed-payment field (DEB-18) as a binary flag only; ignores the credit-register field (DEB-19).
Cap ladder	Partial	Only "resolved→74 / current→39" are applied, and to the overall 8-component score rather than the debt score. Missing: >30d→19, historical→59, payments>50%/60%, debt≥assets, utilisation>100%.

Net: the debt-health score is the documented free-tier approximation (payments-to-income + debt-to-assets + a binary discipline flag). The full analyst engine — BNB benchmarking, utilisation, credit-register history and the full cap ladder — is scoped as paid-tier work and is not yet implemented.

5. Assumptions & notes

- **Annual property tax (EXP-03):** divided by 12 to a monthly figure, per that field's own rule. The cash-flow sheet's shorthand formula only explicitly divides the annual-irregular field; both interpretations are applied consistently and this single point is worth confirming with the analyst.
- **Report presentation vs. numbers:** the cash-flow numbers are faithful, but the prototype's result screen does not yet render the full "Отчет паричен поток" layout (per-component 5-level labels, the monthly overview table, the separate debt-load rating). The headline 0–100 figure is a blended 8-component score; the cash-flow figure verified here is the cash-flow component.
- **Full vs. limited:** identical scoring engine; both were run and returned identical results.

6. How to verify (reproduce the results)

The two check scripts are included in the project folder and require only Node.js. Each loads a prototype, re-computes the methodology independently, and prints the comparison.

Cash-flow (expect: ALL 6 PROFILES MATCH, exit 0)

```
node cashflow-methodology-check.js prototype-full.html
```

```
node cashflow-methodology-check.js prototype-limited.html
```

Debt-health (prints per-component conformance + gaps)

```
node debt-methodology-check.js prototype-full.html
```

```
node debt-methodology-check.js prototype-limited.html
```

The independent implementation of each methodology lives inside the corresponding check script, so the client (or an independent reviewer) can inspect the exact formulas used and confirm they match the spreadsheet.